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Non-local means algorithm with adaptive patch size and bandwidth

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ABSTRACT

Non-local means algorithm is an effective denoising method that consists in some kind of averaging process carried on similar patches in a noisy image. Some internal parameters, such as patch size and bandwidth, strongly influence the performance of non-local means, but with the difficulty of tuning. Many solutions for choosing these two parameters, like cross-validation and Steins unbiased risk estimate criterion, are successful but computationally heavy. In this paper, we introduce a new feature metric that is capable of providing a quantitative measure of geometric structures of image in the presence of noise. The proposed region-based non-local means method first classifies a noisy image into several regions. Then, a local window and a local bandwidth value are selected pixel-wisely according to the property of each region and the local value of the new feature metric. Experiments on standard test images show that the proposed method outperforms the original non-local means version by around 1.34 dB and is comparable to or better than the performance of the current state-of-the-art non-local means based denoising algorithms, both visually and quantitatively.

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1. Introduction

Digital image is a helpful basis and multimedia for us to learn, transmit as well as store information. However, during the image acquisition and transmission process, images are often corrupted by the additive noise, which makes image denoising still an open field study in image processing area. The image denoising problem aims at attenuating noise while preserving the fine structures, textures and details. Over the years, many approaches have been proposed for that purpose, such as the anisotropic filtering (AF), the total variation (TV), and the patch-based filters; their summaries can be found in literatures [1,2]. The AF algorithm [3] implements noise removal and edge enhancement by encouraging diffusion along edges and prohibiting it across edges. However, it fails to recover sharp edges since the solution of its model is in fact smooth [5], meanwhile, the performance on flat areas is even no better than a simple Gaussian blur [1]. TV [4], another popular denoising method, needs to find a suitable prior that enforces the smoothness of the denoised image and achieves detail preservation at the same time. The drawbacks of this approach fall into the following two aspects: (i) it tends to cause stair-casing effect and (ii) small structures in the output image appear to be over-smoothened [1,5].

Both AF and TV fail to handle textured regions effectively that the details in the output images are prone to be washed out. In order to overcome this drawback, a class of so-called patch-based filters have been proposed recently and become exceedingly popular in denoising domain. This kind of filters estimates the intensity of a pixel in the noisy image by some kind of averaging of the intensity values of its neighboring pixels. The very familiar Bilateral Filter (BF) [6] is just one of the earliest examples of such filters, which combines gray levels based on both geometric and photometric distances between a pixel and its neighbors. BF is very simple and easy to calculate, however, it is not able to give a good performance if the noise standard deviation is large [2].

The non-local means (NLM) [1,9–12] algorithm is a simple extension of BF in two aspects: firstly, it replaces the pixel-wise measurement of the photometric term in BF with the one that is patch-wise; secondly, ignoring the spatial distance makes it possible to exploit the local and non-local redundancies existing in the noisy image. Generally speaking, NLM filter performs well for both noise removal and detail preservation. However, the primary weakness of NLM is the rare patch effect, or sometimes refers to the "noise halo". It results from the fact that complex textured regions exhibit lower levels of patch redundancy, a lot of noise would be left in the denoised version as a result [7,8]. Moreover, another drawback of NLM called "patch jittering" blur is the suffering from a noticeable blur effect observed in several parts of the denoised image, which is caused by the large patch size and large bandwidth used in NLM framework [8].

Indeed, it is well known that appropriate choices of some internal parameters in NLM framework, e.g. the patch size and the kernel bandwidth used to control the smoothing effect, would help to reduce these two effects. Specifically, for patch size, Louchet and Moisan [8] point out the blur effect in NLM is aggravated by the

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(a) (b) (c)

Fig. 1. Denoising of Barbara image ($\sigma = 20$) by different bandwidth values. (a) Noisy image; (b) the global bandwidth is 20; (c) the global bandwidth is 12. It is obvious that a larger bandwidth value leads to a better noise removal but poorer details preservation, while a smaller bandwidth value results in excellent texture conservation at the expense of noise retaining.

increase of patch size, leading to the conclusion that a larger patch size is more suitable for smooth areas and a smaller patch size is better for textured areas. As for bandwidth, which is the main task for NLM parameter selecting problem, if it is fixed for the entire image, utilizing a larger one would produce an exaggerate blur (Fig. 1(b)) that results in loss of finer details, while a smaller one may amplify the rare patch problem, that is, the denoising capability of NLM is weakened (Fig. 1(c)) [7,8]. How to optimize only one parameter or both are proposed in several literatures like [9–12]. In [9], Van De Ville and Kocher utilize Steins unbiased risk estimate (SURE) for the minimization of the mean square error (MSE) of the NLM, and globally select the optimized bandwidth afterwards. Later, this idea is extended with the consideration of a linear combination of different estimators [10]. Besides that, Kervrann and Boulanger [11] manage to perform a nearly parameter-free NLM, by setting the bandwidth according to a quantile of a Chi-squared distribution and selecting the patch size through an iterative procedure; moreover, it utilizes a particular metric to compute the patch-based photometric distance, involving local noise variance within each patch.

Though these methodologies are quite successful, they all need iteration procedure and are quite computationally complex [7]. For simplicity, some literatures have proposed a sort of classification-based NLM strategy: firstly, a noisy image is classified into several region types, and then, NLM parameters are adaptively selected accordingly. In [12], it considers separating flat areas from textures in a given image. Subsequently, a smaller patch size is adopted in textured parts, which claims that a tender smoothing operation would be carried out in order to preserve image details as much as possible, comparing to the one imposes on the flat areas. In spite of its performance, neglecting the influence of additive noise makes it unable to perfectly discriminate the flats from textured areas; as a result, any misclassification would deteriorate the visual impression of the output image, because of the entirely different treatment applied to each region type. In order to solve this disadvantage, we propose a new image classification criterion that takes local noise variances into account, and is capable of capturing local geometric structures as well. What is more, unlike [12] only considers choosing patch size, we also address the problem of bandwidth selecting, and the adaptive bandwidth parameter selecting strategy is not only according to the classification result as most classification-based NLM methods do, but also taking account of the texture degree and noise level information of each pixel, in other words, this parameter is selected pixel-wisely.

The remainder of this paper is organized as follows. In Section 2, a general overview of NLM method is presented. Next in Section 3, we describe the proposed method in details. Experiments and comparisons between our approach and several state-of-the-art methods are demonstrated in Section 4. And finally, conclusions are drawn in Section 5, and some perspectives of this work are given as well.

2. The non-local means algorithm

Usually, the measurement model for the denoising problem could be expressed as

$$\mathbf{y} = \mathbf{z} + \mathbf{n} \tag{1}$$

where $\mathbf{z}$ stands for the vector representation of the noise-free image, $\mathbf{n}$ is the additive white Gaussian noise (AWGN) of variance $\sigma^2\mathbf{I}$, and $\mathbf{y}$ is the observed noisy image. The pixel-wise non-local means (NLM) algorithm is an adaptive average filter based on the similar patches in the search region that maps the observed data $\mathbf{y}$ onto $\hat{z}$ as follows:

$$\hat{z}(i) = \sum w(i,j)\mathbf{y}(j), \quad j \in \mathbf{S}(i) \tag{2}$$

where $\mathbf{y}(j)$ denotes the intensity value at position j in the noisy image, $\hat{z}(i)$ is the estimated intensity value at pixel i, and $\mathbf{S}(i)$ represents an $S \times S$ search region around i. $w(i,j)$ is a similarity weight which depends on the patches around i and j respectively, of the following form:

$$w(i,j) = \frac{1}{C(i)} \exp\left\{-\frac{\|P_i - P_j\|_2^2}{p^2h^2}\right\} \tag{3}$$

where $\|\cdot\|_2^2$ is the Euclidean distance, h is the bandwidth which controls image blurring degree, $C(i) = \sum_j w(i,j)$ is a normalized factor, p is the patch width and P_i, which is defined as

$$P_i = \left(\mathbf{y}(i+\tau), \tau \in \left[-\frac{p-1}{2}, \frac{p-1}{2}\right] \times \left[-\frac{p-1}{2}, \frac{p-1}{2}\right]\right) \tag{4}$$

represents a vector (of size $p^2 \times 1$) of intensity values taken from the neighborhood of pixel i.

3. The adaptive NLM procedure

The traditional NLM filter uses fixed but not optimized parameters for all types of patches, which would lead to the noise halo and blur problems, according to the aforementioned analysis. For this reason, most classification-based NLM variants have been proposed to select the parameters according to the region types; specifically, they manage to split an image into two parts: namely, the textures and the flats. However, these variants are not adapted well to the case of noise, as it hardly tells flats from textures corrupted by noise, or textures from the contaminated flats. Because entirely different treatments are carried on varied region types, any misclassification will affect the final image quality greatly. For example, an original flat area is prone to be labeled as the textured one in the presence of noise. Then, the local additive noise contained would be largely left in the corresponding part of the output image, due to the detail preservation strategy applied to the textured regions. In this section, we propose a new metric, capable of capturing local

structures and local noise levels, by fully exploring the property of image structure tensor. The image structure tensor, formed as the outer product of the image gradient matrix with itself, represents the orientation of the image, and is calculated pixel-wisely with the following form:

$$\mathbf{T}(i) = \tilde{G}_i \tilde{G}_i^T = \begin{bmatrix} \sum_{j\in\tilde{W}_i} (g_x(j))^2 & \sum_{j\in\tilde{W}_i} g_x(j)g_y(j) \\ \sum_{j\in\tilde{W}_i} g_x(j)g_y(j) & \sum_{j\in\tilde{W}_i} (g_y(j))^2 \end{bmatrix} \tag{5}$$

where $\mathbf{T}$ denotes the image structure tensor, g_x and g_y are the gradients along horizontal and vertical directions, $\tilde{W}_i$ is an $N \times N$ local analysis window around the pixel of interest i, and $\tilde{G}_i$ is the local image gradient matrix, defined as follows

$$\tilde{G}_i = \begin{bmatrix} \vdots & g_x(j) & \vdots \\ \vdots & g_y(j) & \vdots \end{bmatrix}, \quad j \in \tilde{W}_i \tag{6}$$

As a common feature detection method, the image structure tensor is widely used in corner detection, texture segmentation, image diffusion and optical flow estimation. Our proposed pixel classification criterion stems from the facts that image structure tensor contains local structure information [13], and its eigenvalues $\tilde{s}_{i,1}$, $\tilde{s}_{i,2}$ (in descending order) are increasing with the local noise variance σ_l^2 when the original clean image patch is corrupted by noise [14]. Specifically, the relations between eigenvalues of the noisy and noiseless local windows could be written as

$$\tilde{s}_{i,1} = \sqrt{s_{i,1}^2 + \xi N^2 \sigma_l^2} \quad \text{and} \quad \tilde{s}_{i,2} = \sqrt{s_{i,2}^2 + \xi N^2 \sigma_l^2} \tag{7}$$

where $\tilde{s}_{i,1}$ and $s_{i,1}$ are eigenvalues of the structure tensors which are computed from the noisy and noiseless reference windows respectively, and ξ is a constant related to the filter mask used in image gradient computation (e.g. $\xi = 1/2$, if the filter mask is $(1/2) \times \begin{bmatrix} 0 & 0 & 0 \\ -1 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, (1/2) \times \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$).

Due to the fact that $s_{i,1} \approx s_{i,2}$ in flat regions and $s_{i,1} \gg s_{i,2}$ in texture/edge regions [12,14,15], the relative discrepancy between these two eigenvalues indicates how strongly the distribution of the gradients in a noise free window is biased toward a particular direction. As a matter of fact, some authors like Zeng and Lu [12] and Choi et al. [15] utilize this relative discrepancy as a metric to classify an image into the flat and textured classes. However, this measure is not well adapted to the various additive local noise σ_l^2 as such noise alters the discrepancy in an inconsistent way. To alleviate this problem, we define a metric H as follows:

$$H(i) = \frac{\tilde{s}_{i,1}^2 - \tilde{s}_{i,2}^2}{\tilde{s}_{i,1}^2 + \tilde{s}_{i,2}^2} = \frac{s_{i,1}^2 - s_{i,2}^2}{s_{i,1}^2 + s_{i,2}^2 + 2\xi N^2 \sigma_l^2} \tag{8}$$

Eq. (8) illustrates that in a noisy patch, H is roughly inversely proportional to σ_l^2, and if σ_l is sufficiently large, the value of $H \approx (1/2\xi N^2 \sigma_l^2)$ is driven to zero with increasing σ_l. As mentioned above, $s_{i,1}$ and $s_{i,2}$ represent the strengths of the dominant direction and its perpendicular of the gradients in the original noiseless local window respectively. Hence, the value of H is determined by both the local property of the region (i.e. flat or textured) as well as the local noise level it contains. Considering a noise-free patch with strong geometric structures (i.e. the edged area), the discrepancy between the two eigenvalues is very large, leading to $H \approx 1$. However, the additive noise leads to the reduction of H, in keeping with the fact that such corrupted patch is less structured. Overall, H is able to serve as the criteria of classifying a given noisy image into several classes regarding to the local geometric structures and local noise levels.

Although the metric H behaves well in the presence of noise, it suffers from a shortcoming that a large patch size may cause $2\xi N^2 \sigma_l^2 \gg s_{i,1}^2$ that, even this patch is only slightly corrupted and contains abundant of textures, the resulting H is also near zero under such circumstance that a large patch size is applied. So we need to incorporate another kind of metric, capable of reflecting the strengths of textures, as a refinement to impose the local geometric structures on H. As far as we know, several feature descriptors could be used for texture description, for instance, histogram of oriented gradients (HOG) [16], or local binary pattern (LBP) [17], etc. For simplicity, a first-order histogram based feature F [18] is adopted in this work, with the following expression:

$$F(i) = 1 - \frac{1}{1 + \varsigma_i / 255^2} \tag{9}$$

where ς_i is the second moment of a gray-level cumulative histogram of a region (as the same size of $\tilde{W}_i$, which is fixed as 9×9 in the experiments) centered at i. A higher F value indicates the current pixel is more likely to locate in a texture/edge region, while a lower F value suggests the pixel is more probably at a smooth area. However, it is worth mentioning that any possible texture descriptor could be an alternative, and may lead to a better classification result, since our way of computing F is very crude and the only reason we choose it is that the histogram is quite easy to compute.

Summarizing the aforementioned analysis, we come to a more robust classification metric R by combining H and F together

$$R(i) = \left(1 - \frac{1}{1 + \varsigma_i / 255^2}\right) \frac{\tilde{s}_{i,1}^2 - \tilde{s}_{i,2}^2}{\tilde{s}_{i,1}^2 + \tilde{s}_{i,2}^2} \tag{10}$$

Taking both local geometric structures and local noise variances into consideration, R could simply partition the whole image into four classes, namely, the texture with small σ_l^2 (c_1), the medium part (c_2), the texture with large σ_l^2 (c_3) and the flat area (c_4):

$$i \in \begin{cases} c_1 & NME(i) \geq T_1 \\ c_2 & T_2 < NME(i) \leq T_1 \\ c_3 & T_3 < NME(i) \leq T_2 \\ c_4 & NME(i) \leq T_3 \end{cases} \tag{11}$$

where T_1, T_2 and T_3 are the bin values that are respectively corresponding to 90%, 70% and 30% in the cumulative histogram of R. In Fig. 2, we demonstrate the robustness of classification using R criterion. Even in the presence of Gaussian noise with standard deviation 20, the classification result obtained from the noisy Barbara image (Fig. 2(c)) is slightly distorted that some areas which were viewed previously as smooth regions in the original clear Barbara image (Fig. 2(b), colored in dark blue) are thought of as the textures with large local noise variances [the light blue speckles in Fig. 2(c)]. This is due to the fact that the simple texture descriptor F is sensitive to noise. In the following, we will illustrate how to overcome this misclassification drawback by applying some nonlinear functions in the process of bandwidth selection. In addition, we also present the classification result based on the relative discrepancy of two eigenvalues technique illustrated in [12] in Fig. 2(d), where a lot of details such as eyes are misclassified into the smooth areas.

In the next step, multiple NLM operators – NLM filters with different patch sizes and bandwidth values, are adaptively applied to the noisy image according to the classification results. Before we put forward our rules of parameter selecting, let us first come back to the rare patch and blur artifacts mentioned earlier. The rare patch problem, often observed in some particular image structures, such as corners, edges, and textured parts, is caused by the inability to

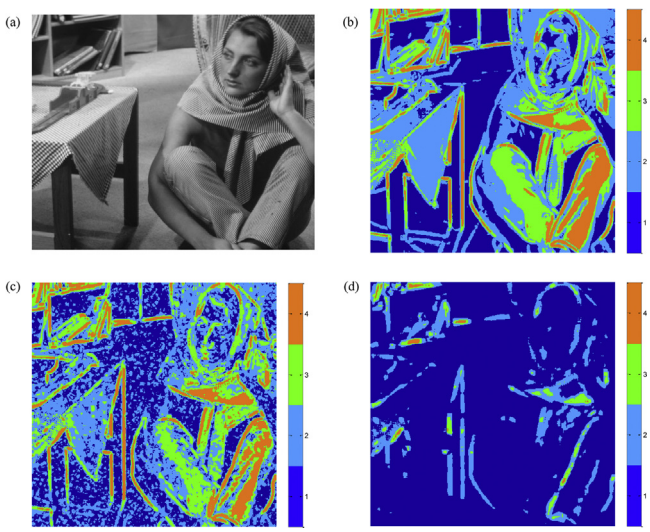

Fig. 2. Classification based on R illustrated on clean and noisy Barbara image of standard deviation 20. (a) Original image; (b) noise-free image classification; (c) noisy image classification; (d) classification result in [12].

find enough patches with similar configuration, and leads to the ineffectively noise suppression. One possible solution is to use a larger bandwidth in textured areas so as to reduce the strength of the photometric distance penalty. On the other hand, for the blur problem, which is mainly caused by the large bandwidth as well as the large patch size used in NLM framework, we opt to apply a smaller patch size in textured areas than in the flats, with the purpose of eliminating blur effect as well as preserving image details.

Specifically, the patch sizes selected for region types c_1, c_2 and c_4 in this paper are 7×7, 9×9 and 13×13, respectively. But c_3 is a little more complicated, owing to the fact that the local noise contained in this region type is larger than other textured areas c_1 and c_2, or sometimes is even overpowered the local textures; then in order to recover the degraded geometric structures, the suitable patch size must be small enough, fixed as 5×5 in all experiments in this paper. In the meanwhile, the scheme for selecting bandwidth is devised as follows: we use a rule illustrated in [11] that $h\propto0.7\sqrt{(\chi^2_{p^2,0.99}/1.529p^2)}\sigma$, where $\chi^2_{p^2,0.99}$ denotes the quantile of a Chi-square distribution with p^2 (relating to patch size) freedom, in order to adjust the bandwidth to different region types. In the proposed method, the values of local bandwidth are chosen as $h\propto0.7\sigma$ for c_1 and $h\propto0.638\sigma$ for c_4, indicating a larger bandwidth for textured areas, as desired. Besides that, we also set the

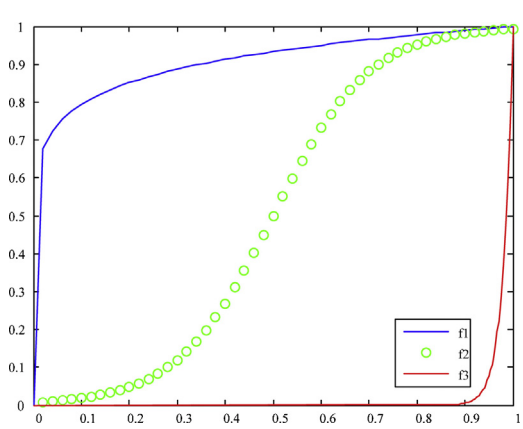

Fig. 3. Diagram of three nonlinear functions.

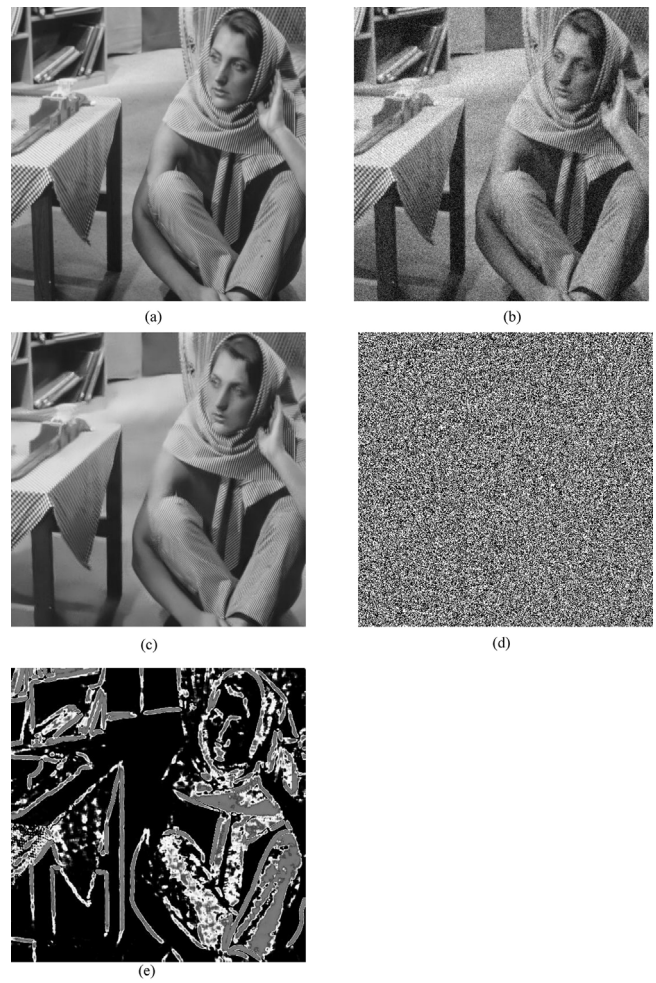

Fig. 4. Decomposition of the noisy Barbara image. (a) Noise-free image; (b) noisy image ($\sigma=25$); (c) denoised image; (d) noise component; (e) bandwidth map associates to the noisy image.

bandwidth proportional to the function of R since it takes into account the local geometric structures, and in this way, the values of bandwidth for the entire image are turned from some discrete values, as most classification-based NLM method do, into a continuous curve, or in other words, forming a bandwidth map. Specifically, if we denote $k_i=0.7\sqrt{(\chi^2_{p_i^2,0.99}/1.529p_i^2)}$, $i=1, 2, 3, 4$ to represent the coefficients calculated from Chi-square distribution, and then, the bandwidth map is defined as

$$h(j)=\begin{cases} f_1(\tilde{R}_1(j))\sigma & \text{if } j\in c_1\\ f_2(\tilde{R}_2(j))\sigma & \text{if } j\in c_2\\ f_3(\tilde{R}_3(j))\sigma & \text{if } j\in c_3\\ k_4\sigma & \text{if } j\in c_4\end{cases}$$

$$\begin{aligned} f_1(x)&=(k_1-k_2)x^{0.1}+k_2\\ f_2(x)&=k_3+\frac{k_2-k_3}{1+\exp(-10x+5)}\\ f_3(x)&=(k_3-k_4)x^5+k_4\end{aligned} \tag{12}$$

where p_1, p_2, p_3 and p_4 are the corresponding patch sizes for each region, and $\tilde{R}_i(j)$ is the normalized value of R which belongs to c_i, $i=1,2,3$ respectively. Fig. 3 illustrates several curves of the nonlinear functions $x^{0.1}$, $(1/1+\exp(-10x+5))$ and x^5 used in Eq. (12). Take

Fig. 5. Comparisons of the visual denoising performances of the proposed approach and other NLM based methodologies on image Barbara with $\sigma = 25$(detail). (a) Noisy image; (b) NLM [1]; (c) ours; (d) NLM-SAP [7]; (e) PCA-SURE-based NLM [10]; (f) RBNLM [12]. The software of NLM-SAP and SURE-based NLM are available at http://josephsalmon.eu/code/index_codes.php?page=NLMSAP and http://miplab.epfl.ch/software/NLM-SURELET.zip respectively.

$f_3()$ as an example, the reason we resort to this function is that, for pixels that are classified into c_3 but with low values of R near T_3, some of them actually belong to c_4 in the original noise-free image and this misclassification is mainly caused by the crude texture descriptor F. For this reason, it is preferable to perform denoising other than details preservation on those pixels, that is, following the requirement for pixels assign to c_4. As a result, we choose the power function x^5 which maintains its stability at first and then increases quickly, to meet our needs. The reasons for employing another two nonlinear functions are similar to the above one: as for $f_2()$, since the treatments for c_2 and c_3 are alike, both requiring detail preservation, then a gradually changed curve would be satisfying, thus we resort to a sigmoid function; on the other hand, patches labeled as region type c_1 are prone to have the noise halo effect, so we employ $x^{0.1}$ as to quickly lift up the bandwidth values as to avoid such artifact.

Fig. 6. Comparisons of the visual denoising performances of the proposed approach and other NLM based methodologies on image Man with $\sigma = 25$(detail). (a) Noisy image; (b) NLM [1]; (c) ours; (d) NLM-SAP [7]; (e) PCA-SURE-based NLM [10]; (f) RBNLM [12].

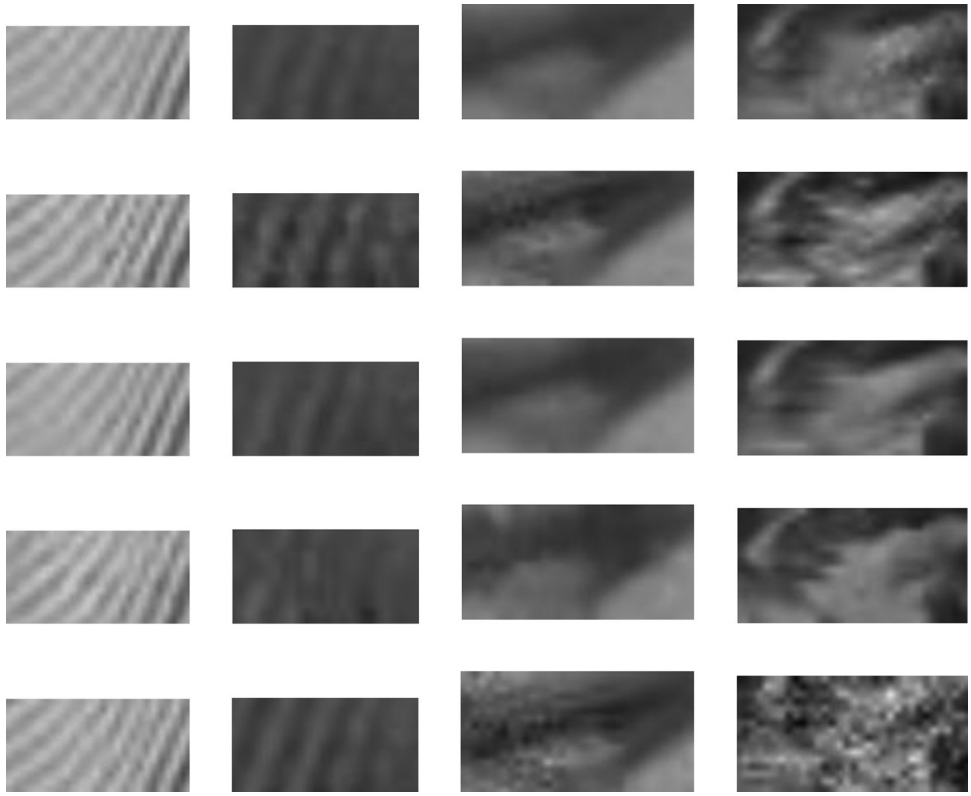

Fig. 7. Details of some parts containing noise halo effect in Figs. 5 and 6 (zoom factor 2). Row 1: NLM [1]; Row 2: ours; Row 3: NLM-SAP [7]; Row 4: PCA-SURE-based NLM [10]; Row 5: RBNLM [12].

4. Experimental results

In the following, we evaluate our method through experiments on some classical images across different noise levels. The application of this method is implemented in MATLAB language. Parameters used in all experiments are listed in Table 1. At first, we show the potential of our method by illustrating with the 512×512 Barbara (Fig. 4(a)) corrupted by AWGN (PSNR = 20.2 dB, $\sigma = 25$) (Fig. 4(b)). The denoised version, exhibited in Fig. 4(c), demonstrates that the noise is reduced in a proper way and fine textures, important features are visually recovered well with little artifact. To better appreciate this, in Fig. 4(d), the residual error image, calculated by means of difference between the noisy image (Fig. 4(b)) and the denoised image (Fig. 4(c)), shows that there are not any noticeable structures in it, indicating the filtered image has kept many details and little blurred. Moreover, we present the bandwidth map associated to the noisy Barbara in Fig. 4(e), where a lighter value corresponds to a larger bandwidth. As expected, pixels with large bandwidth values are located in texture/edge areas and pixels with small bandwidth values correspond to smooth areas, which is a guarantee to avoid the rare patch problem illustrated above. Furthermore, comparing with Fig. 2(c), the misclassification is avoided by the nonlinear functions used in the bandwidth selection, resulting in the extinction of speckles in the smooth areas.

We have also compared our method to various popular denoising methods: NLM method [1], NLM-SAP [7], and PCA-SURE-based NLM [10] and region-based NLM (RBNLM) [12]. The denoised results obtained from these algorithms are compared visually in Figs. 5 and 6, where parts of the image Barbara and the image Man are shown. The PCA-SURE-based NLM, does not manage to perform denoising efficiently that, a noticeable degree of noise remains in the initially smooth areas (cheek, hand in Fig. 5 and cheek, hair in Fig. 6). The NLM and NLM-SAP methods produce better results on this aspect but have the tendency of over-smoothened, which means that both of them suffer from the blur problem. The performance of RBNLM is not stable, for example, in Barbara image, it succeeds in noise removal but is prone of washing out details; while in Man image, it preserves most of details at the cost of noise retaining. The proposed method, on the other hand, is able to find a good compromise between noise removal and details preserving, and avoid the noise halo and blur problems illustrated above. Furthermore, Fig. 7 shows some close-up zones of several parts in Figs. 5 and 6, which correspond to the areas that contain noise halo effect, and it is apparent that our method improves the restoration quality of patches contain such effect.

Moreover, we have also numerically evaluated the algorithms with regard to their capability of restoring the original image. Specifically, we choose peak signal-to-noise ratio (PSNR) and structural similarity (SSIM) [19] as two measures for the restoration quality, defined as:

$$PSNR = 10\log_{10}\left(\frac{255^2}{\sum_i (x(i) - z(i))^2}\right)$$
$$SSIM = \frac{(2\mu_x\mu_z + c_1)(2\sigma_{xz} + c_2)}{(\mu_x^2 + \mu_z^2 + c_1)(\sigma_x^2 + \sigma_z^2 + c_2)} \tag{13}$$

where x denotes the noise-free image, z is the restored version, μ_x and σ_x are average and variance of x respectively, and σ_{xz} is the covariance of x and z. In Table 2, we compare the five algorithms on four classical images: Barbara, Lena, Boat and Man, across four

Table 1
Default parameters used in experiments.

Class	c_1	c_2	c_3	c_4
Patch size $p \times p$	7×7	9×9	5×5	13×13
Search window size $S \times S$			21×21	

Table 2
Comparisons of denoising methods for different noise levels in terms of PSNR (dB) and SSIM (PSNR/SSIM).

σ	20	25	30	35
Image	**Lena** 512×512			
[1]	31.04/0.837	29.92/0.812	28.96/0.789	28.14/0.767
[7]	31.97/0.849	30.79/0.825	29.75/0.802	28.81/0.780
[10]	31.65/0.837	30.51/0.811	29.64/0.787	28.91/0.763
[12]	31.47/0.853	30.94/0.841	30.18/0.823	29.21/0.802
Ours	32.10/0.855	31.14/0.834	30.27/0.812	29.49/0.793
Image	**Barbara** 512×512			
[1]	29.46/0.851	28.11/0.814	26.90/0.778	25.86/0.743
[7]	30.26/0.862	28.87/0.826	27.64/0.790	26.54/0.754
[10]	29.40/0.842	28.08/0.803	27.00/0.765	26.10/0.729
[12]	29.58/0.874	29.00/0.852	28.23/0.825	27.29/0.795
Ours	30.45/0.880	29.42/0.856	28.54/0.830	27.72/0.805
Image	**Boat** 512×512			
[1]	28.54/0.751	27.42/0.714	26.44/0.681	25.58/0.652
[7]	29.58/0.768	28.41/0.730	27.39/0.696	26.48/0.665
[10]	29.56/0.775	28.48/0.737	27.59/0.703	26.85/0.672
[12]	28.84/0.776	28.12/0.745	27.41/0.715	26.61/0.686
Ours	29.94/0.798	28.87/0.762	27.95/0.731	27.10/0.700
Image	**Man** 512×512			
[1]	28.68/0.762	27.60/0.724	26.73/0.692	26.01/0.665
[7]	29.54/0.780	28.46/0.743	27.56/0.710	26.78/0.681
[10]	29.73/0.790	28.68/0.751	27.85/0.716	27.16/0.686
[12]	28.89/0.785	28.18/0.752	27.49/0.721	26.71/0.690
Ours	29.86/0.804	28.82/0.767	27.97/0.730	27.22/0.706

noise levels. It shows that our algorithm exceeds the original NLM greatly, and is comparable to NLM-SAP and PCA-SURE-based NLM in terms of PSNR and SSIM. For some images, although the PSNR gains are marginal, the visual difference can be as obvious as shown in Figs. 5–7. In addition, our method also exceeds RBNLM [12]. From all the experiments, it shows that, both visually and quantitatively, our method makes a good balance between noise removal and detail preservation, and is superior to the state-of-the-art denoising methods.

5. Conclusion

We present an adaptive NLM strategy to solve the rare patch and blur problems occurred in the original NLM. As is well known that, two internal parameters, bandwidth and patch size, are relevant to these problems. The proposed method adjusts these two parameters by making full use of the local noise variances and local geometric structures, which are captured by a new metric advocated in this paper. According to the proposed metric, a classification step is utilized at first to decompose a noisy image into four types of regions. Then, the bandwidth and patch size parameters of NLM, are selected pixel-wisely, regarding to the corresponding metric value and the property of class that each pixel belongs to. Experiment results demonstrate that the proposed method is effective in noise removal and detail preserving, and produce a better performance than some state-of-the-art approaches, both visually and quantitatively. The future improvements for our algorithm could be to resolve the real image-dependent noise problem and realize an automatic parameter selection strategy with respect to the minimization of mean square error (MSE).

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